

Phase 0 Mission Proposal

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Version: MVP Draft

Executive Summary

During launch, the spacecraft is integrated with the selected launch vehicle and remains in launch mode through ascent and separation. During orbit insertion, the launch vehicle injects the spacecraft directly into the mission orbit or into an initial orbit from which the spacecraft reaches the mission orbit using any onboard orbit-adjust capability defined during design. During early orbit operations, the spacecraft enters an initial safe configuration, establishes first contact, acquires attitude knowledge and control, deploys appendages if present, and performs initial health assessment. During commissioning, the spacecraft verifies electrical power, telemetry and telecommand links, attitude control, thermal control, onboard software, payload activation, onboard data storage, payload calibration, and payload data downlink before release to routine service. During nominal operations, the ground segment plans observations, uploads commands, the spacecraft performs nadir imaging over scheduled targets, stores payload data onboard, downlinks telemetry and payload data during ground contacts, and the ground segment processes and delivers imagery products. During contingency operations, the spacecraft autonomously or by ground command enters safe mode, inhibits payload operations, maintains housekeeping telemetry, supports fault isolation and recovery, and returns to the appropriate operational mode after resolution. During end-of-life operations, the spacecraft terminates payload operations, performs final data downlink if planned, passivates applicable stored-energy sources, and executes the approved disposal sequence consistent with the selected debris mitigation standard.

MVP Scope and Limitations

1. This MVP report is generated for a single-satellite mission concept. It does not analyze constellation deployment, multi-satellite phasing, revisit optimization, inter-satellite links, or multi-plane architectures.
2. This MVP report assumes an Earth Observation payload concept. Payload-specific analysis is limited to high-level swath, resolution, mass, and power assumptions where provided.

Mission Overview

- Provide Earth observation imagery from a 550 km circular orbit at 97.6 deg inclination for a mission lifetime of 3.0 years after commissioning.
- Acquire imagery with 1.0 m ground sampling distance or finer at nadir and a swath width of 40 km or greater from the mission orbit.
- Accommodate a payload with allocated mass not greater than 40 kg and nominal operating power demand of 60 W within the mission system architecture.
- Collect, store, downlink, process, and deliver payload data during routine mission operations.

Mission Requirements

ID	Requirement	Rationale
MR-01	The spacecraft shall be designed for operations in a circular orbit with mission altitude of 550 km ± 25 km and inclination of 97.6 deg.	The mission altitude and inclination are selected to provide coverage of the Earth's surface.
MR-02	The mission system shall support an operational lifetime of not less than 3.0 years after commissioning.	Operational lifetime is determined by mission goals and payload selection.
MR-03	The payload subsystem shall provide a nadir ground sampling distance of 1.0 m or finer.	Ground sampling distance is required for the mission's resolution.
MR-04	The payload subsystem shall provide a nadir swath width of 40 km or greater.	Swath width determines the area of coverage per pass.

ID	Requirement	Rationale
MR-05	The payload subsystem mass, including instrument structure, harness, and the mass of the payload, shall be required for with the	And as payload mass is a primary requirement for the spacecraft
MR-06	The electrical power subsystem shall provide at least 60 W of regulated power to the payload according to the payload power	Regulated power to the payload is a primary requirement for the spacecraft
MR-07	The spacecraft shall implement the following operational modes: Defined, early nominal, and end-of-mission	Defined, early nominal, and end-of-mission modes are required for the spacecraft
MR-08	The spacecraft shall accept ground-commanded transitions between operational modes	Centralized orbit and attitude control is required for the spacecraft
MR-09	The spacecraft shall autonomously inhibit payload operations when the payload is in a safe mode	Payload inhibition is required to reduce power and operational risk
MR-10	The spacecraft shall autonomously transition to safe mode upon detection of a predefined fault condition	Detection of a predefined fault condition is required for the spacecraft
MR-11	The spacecraft shall generate and transmit housekeeping telemetry data	The spacecraft shall generate and transmit housekeeping telemetry data
MR-12	The attitude and orbit control subsystem shall support nadir pointing	Nadir pointing is a primary requirement for the spacecraft
MR-13	The spacecraft shall store payload data onboard before transmission	Storage of payload data is required for the spacecraft
MR-14	The communications subsystem shall downlink housekeeping telemetry and payload data	Telemetry and payload data downlink is required for the spacecraft
MR-15	The ground segment shall provide mission planning, command, and data handling	Ground segment support is required for the spacecraft
MR-16	The mission system shall complete commissioning before entry into orbit	Commissioning is required for the spacecraft
MR-17	The commissioning phase shall verify electrical power, communications, and attitude control	Verification of spacecraft systems is required for the spacecraft
MR-18	The mission system shall include an end-of-life disposal sequence	End-of-life disposal is required for the spacecraft
MR-19	The spacecraft shall be passivated at end of life by terminating payload operations	Passivation is required for the spacecraft
MR-20	The mission system shall implement an orbital debris mitigation strategy	Orbital debris mitigation is required for the spacecraft

Mass Budget

Status: GREEN

Subsystem	Mass (kg)	Margin %	Notes
Structure	18.0	20.0	Estimated primary and secondary structure for a small EO space
ADCS	14.4	20.0	Estimated reaction wheels, sensors, magnetorquers, and assoc
Propulsion	7.2	20.0	Assumed minimal propulsion capability for orbit trim, collision av
Communications	9.6	20.0	Estimated TT&C plus payload downlink communications hardw
OBC/Data Handling	6.0	20.0	Estimated onboard computer, mass memory, and data handling
EPS/Power	18.0	20.0	Estimated solar arrays, batteries, power conditioning, and distri
Thermal	7.2	20.0	Estimated spacecraft-bus thermal control hardware excluding p
Payload	48.0	20.0	Payload mass provided as 40 kg in mission context. Applied sta
Harness	6.0	20.0	Estimated spacecraft harness external to payload-mounted har

Power Budget

Status: RED

Mode	Total W
nominal	0.0
peak	0.0
safe mode	0.0

Cost Estimate

Category	Cost (kEUR)	Basis
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Component Alternatives

Category: Fallback

Rank	Product	Vendor	URL	Price	Lead wks
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Warning: Sourcing gap: procurement output unavailable.

Assumptions and Open Items

Assumption: The mission orbit is sun-synchronous or otherwise compatible with maintaining 97.6 deg inclination and 550 km circular altitude for the full operational lifetime without large orbit-change maneuvers.

Assumption: The 1.0 m ground sampling distance requirement and the 40 km swath width requirement both apply at nadir from the mission orbit and to the same payload operating mode.

Assumption: The 40 kg payload mass allocation includes payload-mounted structure, harness, and thermal hardware, and excludes spacecraft bus equipment not mounted with the payload.

Assumption: The 60 W payload power value is the nominal in-operation demand at the spacecraft electrical interface and excludes power conversion losses elsewhere in the bus.

Assumption: The launch service can provide direct injection to the mission orbit or an insertion close enough that any required transfer can be performed within spacecraft mass and delta-v allocations.

Assumption: The mission does not require formation flying, rendezvous, docking, inter-satellite links, or routine off-nadir imaging unless added by later requirements.

Assumption: The mission does not require onboard image processing beyond instrument control, packetization, compression if selected, storage management, and data transmission.

Assumption: A ground segment with sufficient station visibility, spectrum authorization, and data handling capacity will be available to support the required telemetry and payload data return.

Assumption: The spacecraft can satisfy the selected debris mitigation standard at 550 km without a disposal maneuver that dominates spacecraft propellant mass.

Assumption: The radiation, drag, and atomic oxygen environments at 550 km are within the qualification envelope of candidate spacecraft and payload technologies under consideration.

Open question: Which orbital debris mitigation standard will be contractually applicable in Phase A, and what specific post-mission disposal time limit and passivation actions will it impose?

Open question: What spectral bands, radiometric resolution, signal-to-noise ratio, and dynamic range are required for the payload products?

Open question: What image quality metrics in addition to ground sampling distance are required, including modulation transfer function, geolocation accuracy, absolute pointing error, and line-of-sight stability?

Open question: What geographic coverage, revisit time, tasking latency, and percentage of requested targets must be achieved for mission success?

Open question: What average and peak daily payload data volumes are required, and what minimum downlink data rate, contact time, and number of ground stations are needed to close the data budget?

Open question: Is off-nadir imaging required, and if so, what maximum off-nadir angle and what image quality degradation limits are acceptable?

Open question: Will the spacecraft carry propulsion, and if so, what functions are mandatory: orbit trim, drag makeup, collision avoidance, disposal, or all of these?

Open question: What launcher and dispenser environments will bound the design, including separation system, envelope, mass limit, vibration, shock, and initial orbit dispersions?

Open question: What command link and payload data security requirements apply, including encryption, authentication, key management, and export or regulatory constraints?

Open question: What calibration concept is required, including onboard calibration hardware, lunar or stellar calibration, vicarious calibration, and maximum commissioning duration?

Disclaimer

This document is an AI-assisted Phase 0 concept draft. All technical values, cost estimates, procurement options, and requirements must be reviewed and validated by a qualified spacecraft systems engineer before use in design, procurement, proposal submission, or mission decision-making.